CDT Healthcare Data Science

ML for Time Series Module

Practical 01: ECG Signal Processing and Basic Modelling

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0 Lab overview

This lab will go through basic signal processing algorithms, using the ECG as an example.

We will develop an algorithm composed of multiple signal processing steps to locate the characteristic waveforms (P, QRS and T) of an ECG, a task formally termed ECG delineation.

Subsequently, we will compare our algorithm against an existing toolbox, NeuroKit2, on how well the two agree in the beats they detect and in the heart rate and heart rate variability derived from them.

After that, we will use features based on the detected characteristic waveforms to train a classifier that distinguishes different ECG categories (normal rhythm, atrial fibrillation, other rhythms and noisy recordings).

How to work. Open `ECG_Preprocessing.ipynb` and run the cells in order.

Each task asks you to complete one function, described in a box in this note. When you run it, the notebook checks your answer against a reference.

If you have not finished a task, the notebook uses the reference (i.e. the answer to that task) instead, so the rest of the cells still run.

Setup. The lab is in a GitHub repository (https://github.com/junjiechen-chris/CDT-HDS-ECG-Lab-1). You do not need to install anything or use a GPU.

1. Open the notebook in Colab using the badge in its first cell.
2. Save a copy to your Drive, using Save a copy in Drive in the File menu.
3. Run the Setup cell. It downloads the code and data and installs the packages, mainly NeuroKit2 [1] and scikit-learn.
4. Enter the data password when asked. Your tutor will give it out during the lab.

1 What is an ECG signal?

Notebook: Section 1

An electrocardiogram records the voltage between two electrodes placed on the skin. Each normal heartbeat typically consists of a P wave, a QRS complex and a T wave (as shown in the lecture). The interval between two consecutive R peaks is termed the RR interval.

The data are 2,000 single-lead recordings of 9 to 60 seconds at 300 Hz from a handheld device, from the PhysioNet/Computing in Cardiology Challenge 2017 [2, 3]. Each has one expert rhythm label (Table 1). Figure 1 shows one recording from each class.

Label	Rhythm	Recordings	Typical patterns
`N`	normal sinus rhythm	1,187	evenly spaced beats with P waves
`A`	atrial fibrillation (AF)	196	no P waves; irregular RR intervals
`O`	other rhythm	553	e.g. premature (early) beats
`~`	too noisy	64	too noisy to read

Table 1: The rhythm labels.

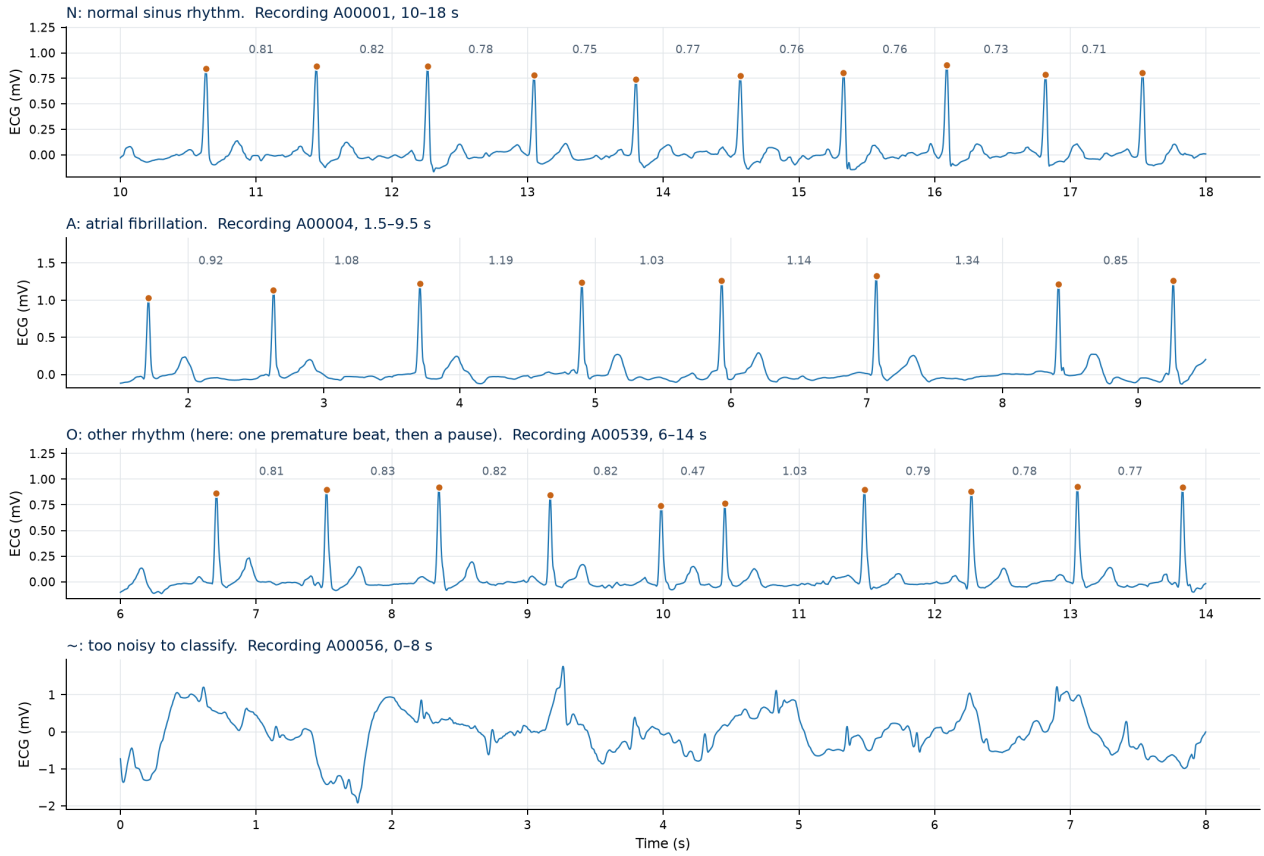


Figure 1: One recording from each class. The dots mark the beats, and the numbers above them give the RR intervals in seconds. `N`: evenly spaced beats, each with a P wave. `A`: intervals that change from one beat to the next, and no apparent P waves. `O`: an abnormal rhythm that is not AF. `~`: no beat can be recognised.

Task 1. Label the P, Q, R, S and T dots on five ECG segments.

2 Measuring rhythm from ECG signals

Notebook: Section 2

This section will look into how to use an established toolbox (NeuroKit2) to locate the R peaks and, based on them, derive heart rate and heart rate variability.

1. **Locate the R peaks.** `nk.ecg_clean` filters the raw ECG, and `nk.ecg_peaks` returns the sample index s_k of each R peak.
2. **Compute the RR intervals.** Convert each index to a time, $t_k = s_k / f_s$ with $f_s = 300\,\text{Hz}$, and take the differences between consecutive beats, $RR_k = t_{k+1} - t_k$. K beats give $N = K - 1$ intervals.
3. **Compute heart rate and heart rate variability** [4]:

$$\text{HR} = \frac{60}{\overline{RR}}, \quad \text{SDRR} = \sqrt{\frac{\sum_{k=1}^{N} \left(RR_k - \overline{RR}\right)^2}{N-1}}, \quad \text{RMSSD} = \sqrt{\frac{\sum_{k=1}^{N-1} \left(RR_{k+1} - RR_k\right)^2}{N-1}}.$$

4. **Compare the feature differences between the four classes.**

Heart rate is similar across the four classes, but RMSSD is much higher in AF than in normal rhythm (median 188 ms against 49 ms), because the RR intervals in AF are irregular (Figure 2).

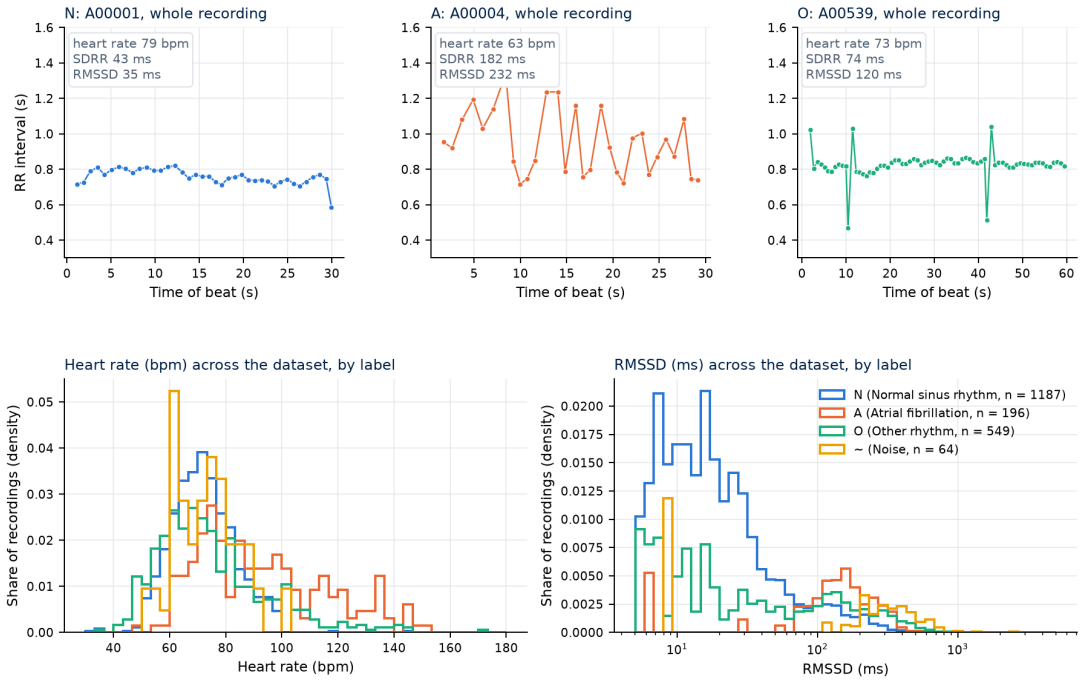


Figure 2: Top: the RR intervals of the `N`, `A` and `O` recordings of Figure 1 over the whole recording, with the heart rate, SDRR and RMSSD calculated from them. Bottom: heart rate and RMSSD (log axis) for every recording in the dataset, by label. Medians: `N` 71 bpm and 49 ms; `A` 89 bpm and 188 ms; `O` 71 bpm and 181 ms; `~` 73 bpm and 394 ms. Beats found by NeuroKit2.

Task 2. Implement `rhythm_features`. Your results are compared with NeuroKit2's `hrv_time`.

Specification: `rhythm_features(rr_s)`

Given `rr_s`, a 1-D NumPy array of RR intervals in seconds, in time order: about 40 values for a 30 s recording, typically between 0.6 and 1.2 s, built from the beats found by NeuroKit2.

You may use NumPy only: the mean, the sample standard deviation, differences between neighbouring values, the square root.

Return a tuple of three floats `(heart_rate_bpm, sdrr_ms, rmssd_ms)`: $60/\overline{RR}$; the sample standard deviation of the intervals in ms; the square root of the mean of the squared successive differences in ms, squared before averaging.

Detector	Task	Agreement	HR (bpm)	RMSSD (ms)
Naive threshold	3	0.71	51.7	319
+ band-pass filter	-	0.72	52.6	288
+ envelope	4A	0.92	6.4	218
+ locate R peak	4B	0.91	5.1	194
+ median imputation	4C	0.90	5.0	146

Table 2: Agreement with the NeuroKit2 reference after each step. Errors are MAE.

3 Locating R peaks from ECG signals

Notebook: Section 3

This section will build a beat detector step by step to locate the R peaks in noisy ECG recordings, following the QRS detector of Pan and Tompkins [5].

1. **Threshold (Task 3).** Use a simple threshold (namely, if the signal is above a threshold, it is considered as one beat), and locate the R peak at its highest point.

2. **Envelope (Task 4A).** Apply a series of operations (differentiation, squaring and averaging) to derive the envelope of the heartbeat area, so that each QRS complex can be detected (Figure 3).

3. **Locate the R peak (Task 4B).** Place each beat at the maximum of the filtered ECG, and keep at most one beat within a certain period.

4. **Median imputation (Task 4C).** Posthoc processing of the detected heart beats.

After each step, we compare the detected beats with those of NeuroKit2 [1] (`ecg_clean`, then `ecg_peaks`) on all 1,996 recordings (Table 2). Beat agreement is $2 \times \text{matched}/(\text{ours}+\text{NeuroKit2})$, with beats matched within ± 100 ms.

> **Note.** NeuroKit2 is a reference, not the ground truth.



Figure 3: The filtered ECG, its difference, the squared difference and the envelope. The envelope has one bump per beat.

Tasks 3 and 4A to 4C. Implement `naive_beats`, `bandpass`, `differentiate` and `envelope`, `locate`, and `correct`, one for each step. Their full specifications are in the notebook.

4 Classifying rhythms with ECG features

Notebook: Section 4

This section will use ECG features to train a classifier that predicts the rhythm label of each recording (`N`, `A`, `O` or `~`), using the PhysioNet/Computing in Cardiology Challenge 2017 dataset [2].

1. **Extract the features.** `extract_feature` uses NeuroKit2 to locate the R peaks (`ecg_clean`, then `ecg_peaks`) and the P, Q, S and T waves (`ecg_delineate`) in each recording, and returns 54 features (Table 3). Heart rate and RMSSD from Section 2 are two of them. The feature table has been computed in advance (`ECG_feature.pkl`).
2. **Split the data.** The recordings are split into training (70%, 1,400 recordings), validation (15%) and test (15%) sets with `random_state=42`, the same split as in Practical 04.
3. **Train a RandomForest** [6] of 100 trees. Each feature is first scaled with a `StandardScaler` fitted on the training set only.
4. **Evaluate on the test set.** The classes are unbalanced, so we use the macro-averaged F1, the mean of the F1 scores of the four classes.

Family	Count	Features
R-peak timing	8	R amplitude (`a_R`); the RR intervals before, after and two after each beat (`RR0`, `RR1`, `RR2`) and their mean (`RRm`); the ratios `RR0/RR1`, `RR2/RR1`, `RRm/RR1`
Wave shape	20	time between wave peaks: `t_PR`, `t_QR`, `t_RS`, `t_RT`, `t_PQ`, `t_PS`, `t_PT`, `t_QS`, `t_QT`, `t_ST`; amplitude difference between wave peaks: `a_PQ`, `a_QR`, `a_RS`, `a_ST`, `a_PS`, `a_PT`, `a_QS`, `a_QT`, and the ratios `a_ST/a_QS`, `a_RS/a_QR`
Heart rate variability	25	**MeanNN** (the heart rate is 60 000/MeanNN), **RMSSD**, and the other 23 time-domain features of `hrv_time`: SDNN, CVNN, MadNN, MCVNN, IQRNN, Prc20NN, Prc80NN, MedianNN, MinNN, MaxNN, SDANN1/2/5, SDNNI1/2/5, SDSD, CVSD, SDRMSSD, pNN50, pNN20, HTI, TINN
Signal quality	1	`SQI`, NeuroKit2's quality index

Table 3: The 54 features of `extract_feature`. Each is computed per beat and averaged over the recording, except the HRV features and the SQI, which are computed over the whole recording. The two features computed in Section 2 are in bold.

Note. For reference, your model is expected to reach an accuracy of 0.79 and a macro F1 of 0.64 on the test set (Table 4).

Class	Precision	Recall	F1	Support
`N` normal sinus rhythm	0.85	0.92	0.88	173
`A` atrial fibrillation	0.71	0.73	0.72	30
`O` other rhythm	0.70	0.63	0.66	86
`~` too noisy	0.67	0.18	0.29	11
Macro average	0.73	0.62	**0.64**	300

Table 4: The reference model on the 300 test recordings.

Task 5. Implement `train_forest` and `predict`. Your test predictions are compared with a reference model and are expected to match it exactly. Save your test macro F1 for Practical 04.

Specification: `train_forest(x_train, y_train)` and `predict(scaler, model, x)`

Given `x_train`, a 2-D float array of 1,400 rows (recordings) × 54 columns (features), unscaled, with ranges that differ by orders of magnitude between columns; `y_train`, an integer array of 1,400 class codes (0 to 3). For `predict`: the two fitted objects and `x`, $n \times 54$ unscaled features.

You may use `StandardScaler` and `RandomForestClassifier`, imported at the top of the cell; their documentation gives the arguments.

Return `train_forest`: `(scaler, model)`, a scaler fitted on `x_train` only (each column centred and scaled by the training set's mean and standard deviation) and a forest of 100 trees with `random_state=42` fitted on the scaled `x_train` and `y_train`. `predict`: an integer array of n predicted class codes, from `x` scaled with the same scaler.

Next: Practical 04. This practical is mainly about a conventional pipeline based on signal processing. Practical 04 will look into training neural networks end to end on the same data and the same split, and compare them with your macro F1.

Submitting your work

Restart the kernel, run all cells to check that your work is reproducible, and save the notebook with your code and plots.

Acknowledgement. The lab is adapted from the CWM Wearable Technology practical by Mauricio Villarroel, with detector material credited to Gari Clifford and Mauricio Villarroel [7]. The 300 Hz implementation is a teaching adaptation, not a numerical reproduction of that detector.

Bibliography

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